A Multi-Agent LLM Framework for Explainable Algorithmic Trading on NSE-Listed Equities

M.Tech Project, Final Evaluation

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Outline

1. Motivation and objectives
2. System design
3. Evaluation protocol
4. Work done since the mid-semester evaluation
5. Results: core and extended universe
6. Explainability evaluation
7. Conclusions, limitations and future work

Motivation

- Most stock-prediction models output a score without a reason, are tested on US indices, and ignore transaction costs.
- A trading desk works differently: technical, fundamental, sentiment and risk specialists give separate views, and these are argued out before a trade.
- Recent papers copy this structure with LLM agents, but there is little evidence on Indian equities under Indian costs.

Objective

Build an explainable multi-agent decision-support system for NSE stocks, and test it out-of-sample with the full Indian delivery cost structure.

Work completed

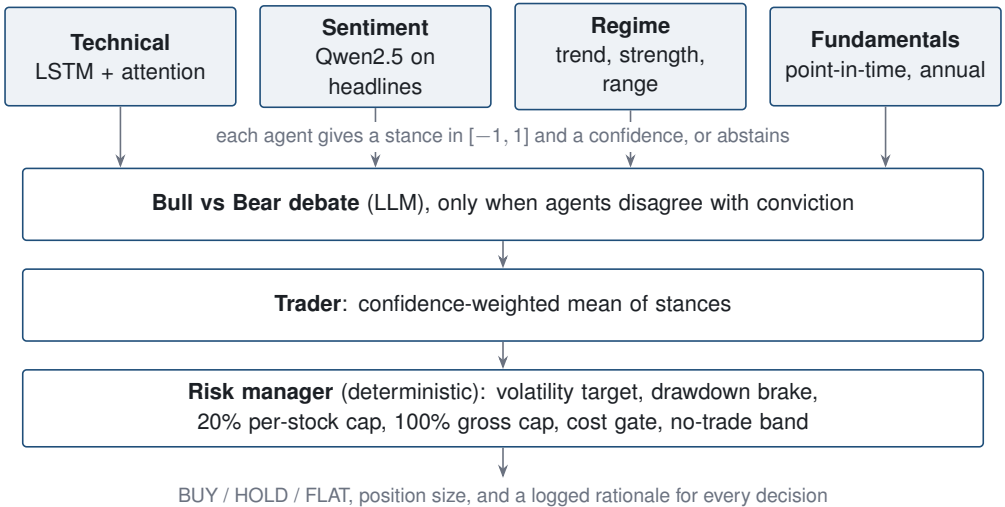
Up to mid-semester

- Technical agent (LSTM with temporal attention)
- LLM sentiment agent on news headlines
- Regime agent
- Bull/bear debate
- Deterministic risk manager
- Cost-aware backtester and baselines

Since mid-semester (November plan)

- Larger LLM (Qwen2.5-3B) compared with 1.5B
- Universe extended from 10 to 30 stocks, longer period
- Sentiment study around earnings days
- Point-in-time fundamentals agent
- Cost-aware sizing and no-trade band
- Full ablation re-run
- Explainability evaluation

System architecture



Evaluation protocol

Validation	Purged, embargoed walk-forward: 3 years train, 6 months test, 10-day embargo, 16 folds
Sample	19,110 out-of-sample decisions, 2018-12-07 to 2026-08-27
Execution	Decide at close of day t, trade at open of day $t + 1$
Costs	STT 0.1% each side, stamp duty, NSE/SEBI charges, GST, 5 bps slippage: about **32 bps per round trip**
Statistics	Block-bootstrap confidence intervals, Holm correction over all configurations, Deflated Sharpe Ratio
New parameters	Fixed before the run: band 0.05, conviction floor 0.10
Reproducibility	Mid-sem configurations reproduce within 0.004 Sharpe; Buy&Hold moved 0.58 to 0.59 after a price-snapshot correction

Fundamentals agent (point-in-time)

- Data: annual P&L and balance sheet for 30 stocks, FY2015 to FY2026.
- A report is used only from FY end + 61 days, following the 60-day filing deadline in SEBI LODR Reg. 33. This avoids look-ahead.
- Features: profit growth, revenue growth, ROE and change in ROE, z-scored across peers on each date.
- Fiscal-year changes are handled separately (HCLTECH 9-month FY2016, NESTLEIND 15-month FY2024).
- Limitations: figures are restated, not as first published; no valuation ratios such as P/E.

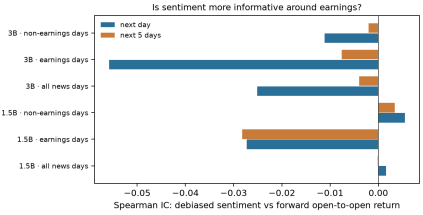
Sample rationale: “FY report Mar 2021 (public since 2021-05-31): return on equity +38.2% is the strongest vs peers (z +2.1)”

Sentiment around earnings announcements

- About 10% of headlines are results news. These days do move more: mean absolute next-day return 133 vs 126 bps ($p = 0.029$).
- On the full corpus, tone does not predict direction at 1 or 5 days.
- Raw tone on earnings headlines shows a reversal (IC about -0.085), but it disappears after debiasing and is smaller than costs.

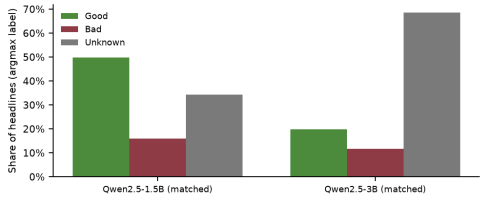
Model	Subset	n	IC	Holm p	Hit rate
Qwen2.5-1.5B	all news	15190	+0.00	1.00	50.1%
Qwen2.5-1.5B	earnings	1803	-0.03	1.00	48.7%
Qwen2.5-1.5B	non-earnings	13387	+0.01	1.00	50.2%
Qwen2.5-3B	all news	6480	-0.03	0.82	49.2%
Qwen2.5-3B	earnings	1803	-0.06	0.36	47.9%
Qwen2.5-3B	non-earnings	4677	-0.01	1.00	49.7%

IC of debiased tone against the next-day return.



Effect of a larger LLM: Qwen2.5-3B vs 1.5B

- The 1.5B model labels Good 3.31 times as often as Bad on the full corpus. This is a bias of the model, not of the news.
- On the same 10,240 headlines and prompt, the two models agree on 60.3% of labels ($\kappa = 0.39$).
- The 3B model is much less bullish and answers Unknown about twice as often.



Model	IC	Holm p	Hit rate
Qwen2.5-1.5B	-0.02	0.13	48.4%
Qwen2.5-3B	-0.03	0.09	49.2%

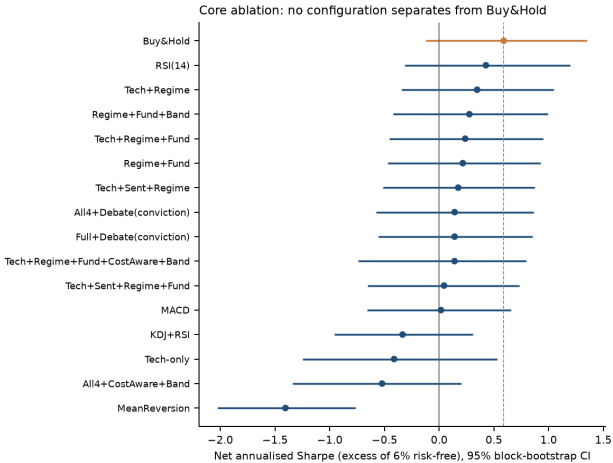
Neither model gives a usable long-only signal.

Ablation results: core universe, net of costs

Strategy	Net Sharpe	Gross Sharpe	Max DD	Turnover/yr	Exposure	Holm p
Buy&Hold	**0.59**	**0.59**	**-38.6%**	**0x**	**100%**	**–**
RSI(14)	0.43	0.47	-39.8%	4x	60%	0.82
Tech+Regime	0.35	0.58	-20.7%	20x	86%	0.82
Regime+Fund+Band	0.28	0.47	-22.2%	16x	84%	0.60
Tech+Regime+Fund	0.24	0.50	-23.2%	23x	88%	0.59
Regime+Fund	0.21	0.48	-23.2%	24x	88%	0.59
Tech+Sent+Regime	0.17	0.66	-25.0%	43x	90%	0.41
All4+Debate(conviction)	0.14	0.60	-24.7%	43x	93%	0.09
Full+Debate(conviction)	0.14	0.62	-27.2%	44x	91%	0.21
Tech+Regime+Fund+CostAware+Band	0.14	0.37	-5.3%	5x	7%	0.82
Tech+Sent+Regime+Fund	0.04	0.59	-25.0%	49x	91%	0.14
MACD	0.02	0.46	-29.9%	43x	78%	0.39
KDJ+RSI	-0.34	0.24	-39.3%	51x	34%	0.01
Tech-only	-0.41	-0.27	-25.4%	5x	16%	0.09
All4+CostAware+Band	-0.52	0.20	-5.1%	14x	7%	0.21
MeanReversion	-1.41	0.68	-79.6%	223x	76%	0.00

10 stocks. Sharpe is annualised, in excess of a 6% risk-free rate. Holm p: paired test against Buy&Hold.

Net Sharpe with 95% confidence intervals



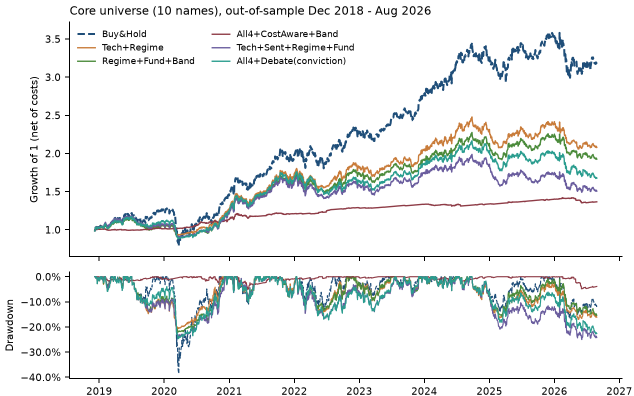
- Every interval overlaps Buy&Hold (0.59).
- Best agent configuration: Tech+Regime at 0.35.
- No configuration is significantly better after Holm correction.
- Mean reversion is significantly worse, because of turnover.

Effect of each component added since mid-sem

Component	Net Sharpe	Observation
Fundamentals agent	Tech+Regime 0.35 to 0.24	Annual rankings add no daily information
No-trade band	Regime+Fund 0.21 to 0.28	Turnover falls from 24x to 16x a year
Edge-vs-cost gate	All4+CostAware+Band -0.52	Few signals clear 32 bps; average exposure only 7%
Conviction-gated debate	0.14 (3 agents), 0.14 (4 agents)	Debate on 10.5% of days instead of 97.4%; result unchanged

The no-trade band is the only addition that improves the net result, and it does so by trading less.

Equity curves and drawdown: core universe

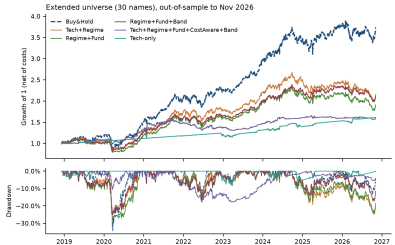


- Buy&Hold ends highest, with a maximum drawdown of -38.6%.
- Tech+Regime limits the drawdown to -20.7%.
- The risk overlay gives up return in rising markets and protects capital in the 2020 crash.

Extended universe: 30 stocks, data to 2026-11-04

Strategy	Sharpe	Max DD	Turnover
Buy&Hold	**0.74**	**-34.3%**	**0x**
RSI(14)	0.66	-38.5%	9x
Regime+Fund+Band	0.36	-25.1%	16x
Tech+Regime	0.35	-24.7%	29x
Tech+Regime+Fund	0.25	-27.8%	29x
Regime+Fund	0.25	-27.8%	29x
MACD	0.23	-29.1%	70x
Tech-only	0.09	-7.7%	7x
Tech+Regime+Fund+CostAware+Band	0.06	-17.8%	17x
KDJ+RSI	-0.92	-64.2%	119x
MeanReversion	-1.97	-91.8%	311x

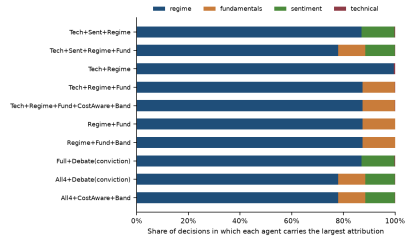
Same conclusion with three times the stocks and on data not available at mid-sem.



Explainability evaluation

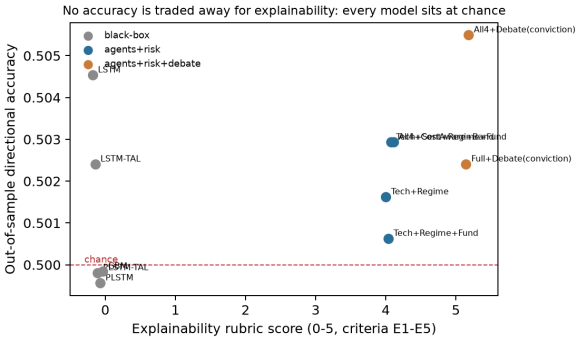
Check	Question	Result
Provenance	Does the rationale name the model that produced the number?	0% at mid-sem, 100% after fix
Integrity	Do stated stances equal logged stances?	100%
Reconstruction	Can the score be recomputed from the opinions?	100%
Deletion	Does removing the top-attributed agent change the decision?	37% to 71% flips (other agent: 0.0% to 3.4%)

The provenance audit found that all 19,110 mid-sem rationales named PLSTM-TAL for LSTM-TAL predictions.



The regime agent decides most calls: the system behaves as a trend filter.

Accuracy vs explainability



- Black-box models (LSTM variants, GBM) are at chance level.
- The explainable agent systems are also at chance level.
- With this little signal, accuracy is limited by the data and not by interpretability.
- The explainable system has one advantage: it shows why there is no edge.

Conclusions

- The full system was built and tested end to end: four agents, debate, risk manager, cost-aware backtester, a paper-trading demo and over 350 automated tests.
- **No configuration beats Buy-and-Hold after Indian transaction costs**, on 10 stocks, on 30 stocks, or on new data.
- None of the November additions changes this result.
- The regime-aware risk overlay reduces maximum drawdown by a third to a half.
- The explanations are faithful by measurement, and the audit found a real provenance bug.

Limitations and future work

Limitations

- Only open 1.5B and 3B language models
- Survivorship bias in the stock universe
- Restated fundamentals
- Statistical power: detectable Sharpe is about 1.0

Future work

- Event-time and intraday horizons
- Vendor point-in-time fundamentals
- Debate with a frontier LLM
- Market-neutral formulation

Thank you

Questions?

Demo (runs in the browser)	`vipluv01.github.io/nse-multi-agent-trading`
Code	`github.com/Vipluv01/nse-multi-agent-trading`